Speculative Commits on Streaming ASR Partials: Measuring Wasted-Action Rates for Voice Agents

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Abstract

Streaming recognizers emit revisable partial transcripts within about a hundred milliseconds before committing a final transcript. Voice agents are increasingly encouraged to act on these partials, but no published measurement describes what happens when the acted-on partial is later revised. We present an engine-agnostic mechanism study of this speculative-action problem. From 360 scripted utterances with ground-truth tool calls, we replay audio through two offline streaming setups (Vosk partial hypotheses; a faster-whisper growing-buffer decoder) and simulate three agent policies: wait-for-final, stability-thresholded speculative commit, and commit-with-rollback. On 290 analyzable utterances (70 excluded for small-model transcription failures; all rates are conditional on the engine recovering ground truth), a stability-thresholded speculative policy (K=2) is waste-free (0%) on direct commands and negations, wastes 93.4% of commits on mid-sentence self-corrections and 72.7% on late qualifiers, for a pooled 38.6% (95% CI 33.4–44.1%) while saving 897 ms per utterance over wait-for-final. Incompleteness — the correction arriving after the commit — dominates recognizer instability as the cause (∼7:1), which is why raising the stability threshold prices latency without buying reliability. All audio is synthesized locally; no human subjects, no cloud services, no paid APIs. We report threshold operating curves, a field-level error profile, and a cause decomposition separating automatic speech recognition (ASR) instability from late-arriving corrections.

1 Introduction

Streaming speech-to-text is now sold as agent infrastructure: Microsoft's MAI-Transcribe-2-Streaming (released 1 October 2026) emits first partials just over 100 ms after audio arrives, so an agent can "start reasoning or calling tools mid-sentence" (MarkTechPost, 2026), yet vendor evaluations report only word error rate and latency (MarkTechPost, 2026). Missing is the failure mode this invites: an agent committing a tool call to a partial the recognizer later revises has taken a *wasted action* before the speaker finished. A blog post frames this as the central governance question for voice agents while noting no hands-on measurement exists (Reed, 2026). We provide it.

We ask: for scripted dialogues with known ground-truth tool calls, what is the wasted-action rate under wait-for-final, stability-thresholded speculative commit, and commit-with-rollback policies, and how much latency does speculation save? Using only free, offline components, we synthesize 360 utterances, replay them through two streaming decoders with genuinely different partial semantics, parse every partial with a validated deterministic parser, and simulate the three policies with bootstrap confidence intervals. Our contributions: (1) the first empirical characterization of speculative-commit waste in streaming-ASR agents we are aware of; (2) an open, reproducible harness (code, corpus, logs) built with no paid service; (3) threshold operating curves for pricing the latency–reliability trade-off before deployment.

2 Related work

Streaming ASR and partial hypotheses. Production streaming recognizers use transducers (Graves, 2012) or adapted offline models. Macháček et al. made the partial/final distinction explicit for Whisper-class models: a transcript prefix is *committed* once n consecutive re-decodes of a growing buffer agree on it (LocalAgreement-n), at 3.3 s latency on long-form speech (Macháček et al., 2023). Commercial APIs expose the same interim/final contract; we study

the contract, not any vendor's numbers.

Agentic speech recognition. The closest prior work, Jiang et al. (2026), studies transcript *quality* for downstream agents via an ASR–refiner architecture; it does not measure what happens when an agent *acts* on a partial before the revision. Zou et al. (2026) proposes a Listen-Think-Speak framework whose foreground Speaker speculates on streaming prefixes while a background Thinker maintains state, built to avoid wasting computation on predictions that must be rolled back; it optimizes an architecture for streaming robustness (with a Pause-and-Repair disfluency benchmark), whereas we measure the wasted-action rate as a metric. The two are complementary: we take revisable partials as given and quantify the cost of acting on them.

Self-correction in spontaneous speech. Mid-utterance repairs ("book a flight to Boston— actually, make that Saturday") are a core phenomenon of spontaneous speech (Shriberg, 1994): the speaker's intent genuinely changes after the agent could have acted, so our corpus is stratified to isolate them.

Tool-using agents. ReAct-style agents interleave reasoning with tool calls (Yao et al., 2023); voice front-ends move the observation stream earlier, into the partial transcript. Speculative execution is well understood in computer architecture, but we know of no published transfer of the idea — or its waste metric — to streaming-ASR agents.

3 Method

3.1 Corpus

We scripted 360 utterances, each paired with a ground-truth tool call (intent plus slots), in six categories: (A) baseline direct commands (60); (B) mid-sentence corrections (90); (C) negations (60); (D) late qualifiers (60); (E) entity-dense requests (45); (F) confusable numbers (45). Slot vocabularies (6 destinations, 7 days, 2 accounts, 6 names, 6 times, 14 amounts) are fixed and the generator is seeded, so the corpus is exactly reproducible. Audio was synthesized locally with Piper text-to-speech (TTS) (`en_US-lessac-medium`) at 22.05 kHz, resampled to 16 kHz mono (mean length 3.31 s); no human speakers, no cloud TTS.

3.2 Streaming engines

Engine 1 – Vosk. Each waveform is streamed through `vosk-model-small-en-us-0.15` in 200 ms chunks; every partial hypothesis and the final result are logged (Alpha Cephei Inc.). Partials are genuine lattice prefixes with non-prefix revisions (verified on pilots).

Engine 2 – faster-whisper growing buffer. As an independent mechanism check on a stratified 120-utterance subset (20 per category), the growing buffer $[0, t]$ is re-decoded every 750 ms with faster-whisper `tiny.en` (int8, CPU, greedy) (SYSTRAN; Radford et al., 2023), each decode emitted as the current unstable partial in the spirit of Macháček et al. (2023)'s LocalAgreement policy. This engine revises far more aggressively than Vosk, giving two distinct points in the design space. Absolute faster-whisper latency numbers are artifacts of the 750 ms re-decode grid, not true streaming latencies.

3.3 Action parser

A deterministic fuzzy parser maps any transcript prefix to a tool call (intent + slots) or *incomplete*: difflib fuzzy-matching of slot vocabularies (token threshold 0.82), last-non-negated mention semantics, mandatory-slot requirements. On the 360 clean template texts it recovers ground truth on all 360; on engine finals, 290 of 360 (Vosk; 70 exclusions, 19.4%, small-model proper-noun errors listed in the released logs) and 107 of 120 (faster-whisper subset; 13 exclusions are tiny.en output-format residuals and proper-noun errors). Utterances whose parsed final differs from ground truth are excluded from policy metrics; exclusion rates are reported, not hidden. The exclusion concentrates brutally: 29 of the 70 exclusions are entity-dense (E), so 64% of that category is dropped and the "0% waste on E" rests on 16 survivors. We bound the bias by running the P2 ($K{=}2$) commit rule on the 70 excluded utterances with ground truth as reference: the policy still committed speculatively on 44, of which 40 (91%) were wasted (the 4 correct commits were all negations); the other 26 fell back to the final, wrong by construction. All headline rates are therefore conditional on "the engine recovered ground truth" and are likely lower bounds on deployment waste. The parser stands in for the agent's NLU; we return to this substitution in §5.

3.4 Policies and metrics

For each utterance, we have a hypothesis timeline $(t_1, x_1), \ldots, (t_m, x_m{=}x_{\text{final}})$. Three policies:

- **P1, wait-for-final:** commit parse(x_{final}) at t_m. Waste-free by construction *given the inclusion criterion* (included finals parse to ground truth by definition), which flatters P1 against deployments where finals can be wrong; the latency baseline.
- **P2, stability-thresholded speculative commit:** commit at the first t_i where parse(x_i) is complete and identical over the last K consecutive hypotheses (K is an engine-agnostic confidence proxy: Vosk partials expose no per-word confidences, but temporal agreement is available on every streaming engine). If no hypothesis qualifies, fall back to the final.
- **P3, commit-with-rollback:** P2's commit rule, then every later hypothesis parsing to a different complete action triggers a rollback plus re-commit.

Metrics: *wasted-action rate* (WAR) = fraction of utterances with committed $\neq$ ground truth; *false-commit cause* — for each wasted P2 commit, *incompleteness* if the commit-time partial is a word-prefix of the final (the correction had not arrived yet) vs. *instability* otherwise; *latency saved* $= t_m - t_{\text{commit}}$; for P3, actions per utterance and final correctness. Paired bootstrap 95% CIs ($B{=}2000$); $K \in \{1, \ldots, 5\}$ swept for operating curves.

4 Results

Of 360 utterances, 290 are included in policy metrics; 70 are excluded (19.4%) because the final transcript does not parse to the ground-truth call — mostly small-model proper-noun errors (29 of the 70 are entity-dense). Mean utterance length is 3.31 s. Non-prefix partial→final revisions occur in 213/290 (73%) of included utterances: revisable partials are the norm, not the exception.

4.1 Operating curves

Fig. 1 gives the operating curves over K. On Vosk ($n{=}290$), at $K{=}2$, the speculative policy commits early on 94% of utterances, saving a mean of 897 ms (95% CI 803–995; median 600 ms) over wait-for-final, but 38.6% of commits are wasted (95% CI 33.4–44.1%). Raising K from 1 to 5

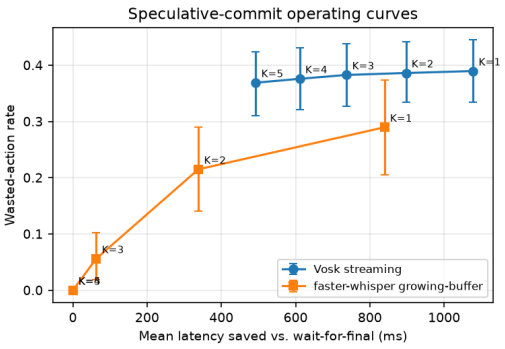


Figure 1: Wasted-action rate vs. mean latency saved, sweeping K: Vosk (flat) prices latency without buying reliability; faster-whisper (steep) WAR falls only as speculation collapses. Error bars: paired-bootstrap 95% CIs.

halves mean latency saved (1078 ms → 492 ms) while the waste rate barely moves (39.0% → 36.9%, overlapping CIs): the threshold prices latency without buying reliability, because 88–101 wasted commits are *incompleteness* versus 10–19 from *instability* — no threshold recovers information the speaker has not yet produced.

4.2 Stratification by utterance category

The pooled rate hides a sharp split (Fig. 2, P2 $K{=}2$). Baseline direct commands (A, $n{=}47$) are waste-free (WAR 0.000, saving 153 ms); negations (C, $n{=}51$) likewise (WAR 0.000). Entity-dense requests (E, $n{=}16$) show no waste either, though proper-noun errors exclude 29 of the 45 utterances in this category, so this zero rests on filtered survivors (see §3.3). Confusable numbers (F, $n{=}45$) cost little (WAR 0.022, one instability waste), vanishing at $K{=}5$ — the one case where the threshold works as intended.

The catastrophic categories are self-correction and late qualification. Mid-sentence corrections (B, $n{=}76$) waste 93.4% of speculative commits (95% CI 88.2–98.7%), saving 1632 ms: the agent books "denver on saturday" a full two seconds before the user says "actually make that seattle on friday." Late qualifiers (D, $n{=}55$) waste 72.7% (95% CI 60.0–85.5%). In both, incompleteness dominates (B: 60 vs. 11; D: 38 vs. 2): the agent acts on a complete-looking parse of a prefix, and the disambiguating words arrive later.

4.3 Field-level error profile

Among wasted P2 commits at $K{=}2$ with the correct intent, the most frequently wrong slots are *day*

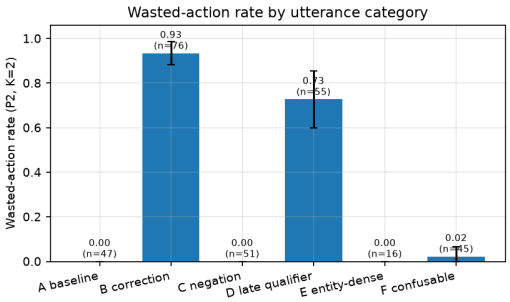


Figure 2: WAR by utterance category (Vosk, P2, $K{=}2$) with 95% CIs: safe on direct commands, negations, entity-dense requests; near-certain waste on self-corrections and late qualifiers.

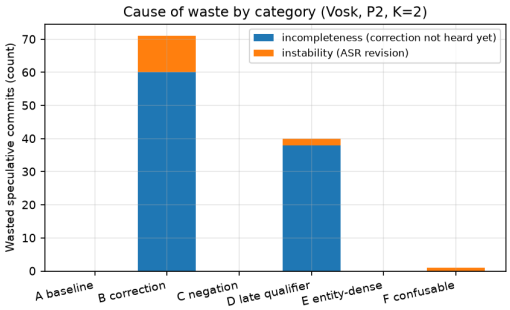


Figure 3: Cause of wasted speculative commits by category (Vosk, P2, $K{=}2$): incompleteness dominates in every category with non-trivial waste.

(34), *account* (20), *dest* (19), *recipient* (16), and *tod* (15) — exactly the slots that self-corrections and late qualifiers target. Amount is almost never wrong (1): stated once, early, rarely revised.

4.4 Commit-with-rollback

P3 issues a mean of 1.54 actions per utterance at $K{=}2$ (1.56 at $K{=}1$, 1.47 at $K{=}5$): the first commit is wasted at the P2 rate, and a rollback+recommit follows once the revision arrives. Rollback restores the outcome by construction, but it cannot un-send the first action — for irreversible tool calls the waste is already realized. On faster-whisper the picture is similar: P3 averages 1.23 actions per utterance at $K{=}2$, the first commit wasted at the P2 rate (21.5%).

5 Discussion

What the curves say. The finding to lead with is the decomposition, not the pooled 38.6% (a secondary statistic on an adversarially stratified corpus): speculation is essentially safe on direct commands and exclusion-style negations, and essentially unsafe on self-corrections and late qualifiers,

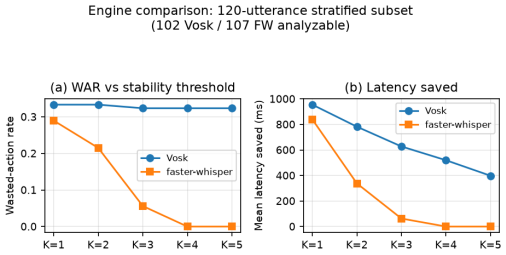


Figure 4: Engine comparison on the stratified subset: (a) WAR and (b) mean latency saved vs. stability threshold K.

with K controlling latency far more than reliability. The policy need not be a single global K: waste concentrates in identifiable constructions (self-corrections, late qualifiers) targeting identifiable slots (day, dest, recipient). A pragmatic design commits speculatively on direct commands but holds the late-mutable slots. The commit-time partial contains no repair marker, so predicting incompleteness waste from the prefix alone is out of reach; slot-level caution is the actionable lever.

Engine comparison. The second engine changes the shape of the trade-off, not just its level (Fig. 4, the stratified 120-utterance subset; per-engine inclusion yields 102 analyzable Vosk vs. 107 faster-whisper utterances). Vosk's lattice partials are stable: WAR is flat in K (0.333 at $K{=}1 \rightarrow$ 0.324 at $K{=}5$) while latency saved falls (955 ms $\rightarrow$ 398 ms) — the threshold prices latency without buying reliability, because waste is incompleteness-dominated. The faster-whisper re-decoder is far less stable: WAR falls steeply (0.290 $\rightarrow$ 0.000) only because speculation collapses (fraction 1.00 $\rightarrow$ 0.075; mean saved 839 ms $\rightarrow$ 0 ms). Here instability causes a larger share of waste (13 of 23 at $K{=}2$), so the threshold bites by refusing to commit at all. The lesson is engine-dependent: with stabilized partials (the commercial-API norm), tune the *policy* (which slots to hold); with unstabilized re-decodes, tune the *threshold*, knowing it trades latency for silence rather than for safety.

Limitations

Five caveats bound the claims. (1) All speech is single-voice synthetic TTS; real spontaneous speech has more disfluency and acoustic variability, so our revision rates are a lower-bound probe of the mechanism, not a deployment forecast. (2) The deterministic parser is a stand-in for an

LLM-based agent NLU; a stronger NLU would parse more partials (more speculation opportunities) but also recover from more revisions. (3) We study English only, with a small slot vocabulary; open-vocabulary tool calls widen the space of possible waste. (4) Vosk's small model and `tiny.en` are weaker than production engines; stronger engines revise less, which moves points along our curves rather than invalidating them. (5) The Piper TTS synthesizer (Piper Contributors) is stochastic across runs (no seed exposed), so re-synthesized audio differs sample-wise; the corpus texts are exactly reproducible from the seeded generator, but the audio is not. All results use the canonical frozen audio set shipped with the release, and a clean-room re-synthesis reproduced every headline within the reported 95% CIs, so the findings are stable and not an artifact of one TTS draw. The harness is engine-agnostic by design so these substitutions can be re-run.

6 Conclusion

Streaming ASR partials invite acting early. We measured what the invitation costs: speculative commit at $K=2$ is waste-free (0%) on direct commands and negations but wastes 93% of actions on self-corrections and 73% on late qualifiers — 38.6% pooled over 290 analyzable utterances to save 897 ms per utterance. Because incompleteness, not recognizer instability, dominates, the fix is not a better threshold but a policy that respects what the speaker has not yet said: hold the late-mutable slots, and never treat a complete-looking prefix as a complete intent. The harness, canonical audio, corpus, logs, and a one-command reproduction script are packaged in the public release accompanying this paper (Zenodo DOI: 10.5281/zenodo.23354707).

Statement of AI Use

The author used the AI tool Muse (Muse Spark 1.3, Meta) for drafting assistance, code scaffolding, and grammatical editing. The contents are reviewed, edited, and attested for accuracy by the author.

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